3 Complex Analysis 2024

Tutorial 5

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Elementary functions

Problem 20 We define the exponential function by $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$, $z \in \mathbb{C}$, and

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Prove the following properties:

(a) $\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$, $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$, $z \in \mathbb{C}$,

(b) $(\cos z)' = -\sin z$, $(\sin z)' = \cos z$,

(c) $e^z = 1 \iff z = 2\pi k i$, $k \in \mathbb{Z}$,

(d) $\cos z = 0 \iff z = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$,

(e) $\sin z = 0 \iff z = \pi k$, $k \in \mathbb{Z}$.

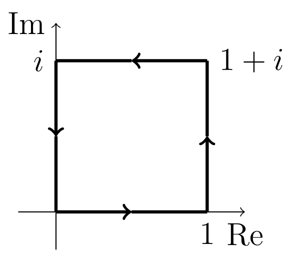
Problem 21 We define $\tan z = \frac{\sin z}{\cos z}$ for all $z \in \mathbb{C}$ with $\cos z \neq 0$. Solve the equations

(a) $\tan z = 1$, (b) $\tan z = i$.

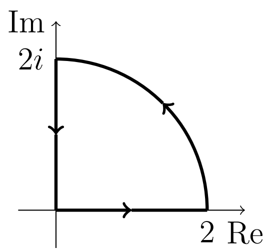
Integrals of complex functions

Problem 22 Evaluate the following integrals by first parametrizing the indicated path:

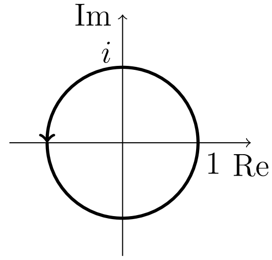
(a) $\int_\gamma \overline{z}\, dz$



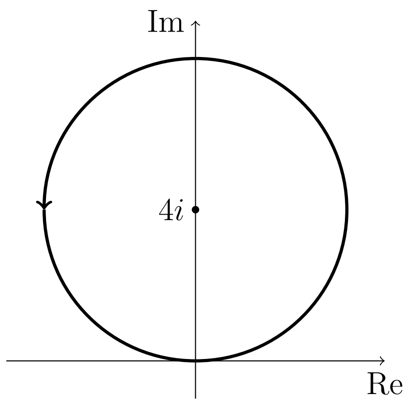
(b) $\int_\gamma \overline{z}\, dz$



(c) $\int_\gamma (1/\overline{z})\, dz$



(d) $\int_\gamma \frac{2z+3}{z-4i}\, dz$



Problem 23 Evaluate the following integrals:

(a) $\int_\gamma |z|^4\, dz$, where $\gamma(t) = i + t$, $t \in [-1, 1]$,

(b) $\int_\gamma z^2\, dz$, where $\gamma(t) = e^{it}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

(c) $\int_\gamma \operatorname{Re} z\, dz$, where $\gamma(t) = t + it^2$, $t \in [0, 1]$,

(d) $\int_\gamma \frac{1}{z}\, dz$, where $\gamma(t) = e^{it}$, $t \in [0, 8\pi]$,

(e) $\int_\gamma e^z\, dz$, where $\gamma(t) = \begin{cases} t, & t \in [0,1], \\ 1 + i(t-1), & t \in [1,2], \\ 3 + i - t, & t \in [2,3]. \end{cases}$